

Fig 4: Input scan values

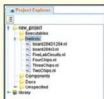


Fig 5: Opening netlist

Installing goJTAG

goJTAG can be installed in MS Windows by double clicking onto the installer file (included in the DVD). Also, for java platforms, you can extract the compressed folder and open the executable jar file with the Java Runtime Environment.



Fig 6: Signal watcher

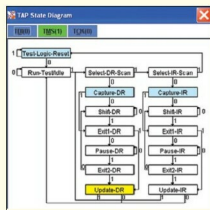


Fig 7: TAP State diagram



Fig 8: Test constructor

sponding netlist.

Boundary scan operations. To start performing boundary scan related operations, you need to switch to the Debug mode, which will open a Test Constructor panel in the left.

In this panel, you can specify instruction and test patterns for each boundary scan device. After specifying the instruction and the test patterns, press the Run button to start performing scan. Additionally, you can also perform scan IR and scan DR operations separately by pressing the corresponding buttons. The results of scans are displayed in the Diagnostic Results panel at the bottom.

Note: You can select the instructions and input scan values directly on the board by simple right clicks and selecting the desired register.

TAP controller. The software also allows you to control the TAP controller of a boundary scan device directly by visualising its state machine. Select Window→Tap State Diagram menu or press the Tap State Diagram button at the toolbar to display the TAP State Diagram.

You can click TDI and TMS buttons to change input signals. Pressing the TCK button, a clock cycle is generated and the TAP controller might change

its state depending on the value of TMS. If you hold Shift key and then press the TCK, it will generate clock edges.

Also, you can speed up your work with TAP controller by clicking on the highlighted states. In this case, the software will generate the required sequence of TCK/TMS signals automatically. Signal Watcher panel at the bottom shows how signals and TAP states change.

Fault injection

The software features an option to inject a fault in your design and train yourself in identifying the fault by means of boundary scan. You can inject the following fault types:

1. Opens (stuck-at model)
2. Shorts (Wired-OR, Wired-AND or Dominant models)

To inject a fault, select Training→Inject Fault... option. In the opened dialog box, you can select fault type and location or inject any random fault. Now, select Training→Check Fault option to check the injected fault using various combinations. If you want to remove the injected fault from the board, select Training→Remove Injected Fault option.

goJTAG is a very handy JTAG/boundary scan tool which has all the features to provide you with a clear illustration of JTAG test principles. The software gives you the option to connect a hardware and control it utilising a boundary scan controller. You can look inside the boundary scan cells, walk through TAP controller states, build your own programs and execute SVFs, import your boundary scan description language (BSDI) files and even create a representation of your own board designs and stimulate boundary scan on it.

Not only you can learn the concepts of JTAG/boundary scan but it also allows you to inject fault and observe its effect(s) on the design, fine-tuning your learning and understanding of the testing of boards. ●

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